

Genes
Tell the Story
We Decode It

Gene Level Quantification and Differential Gene Expression Analysis

From Aligned Reads to Biological Insights

RNA-seq Workshop | Bioinformatics & Computational Biology

Expression
Function
Pathways
Biomarkers
Precision Medicine

Aligned Reads (BAM)



Sorted & indexed
BAM files

Gene Quantification (featureCounts)



Count reads per gene
using annotation (GTF)

Count Matrix

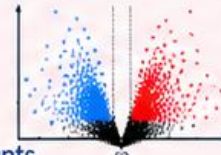
Gene	C1	C2	C3	T1	T2	T3
Gene1	120	95	102	30	28	25
Gene2	0	4	3	50	48	52
Gene3	560	620	580	210	205	198
...	...	...	...	...	...	...

Matrix of raw counts
(genes $\times$ samples)

Differential Expression (DESeq2 / edgeR)



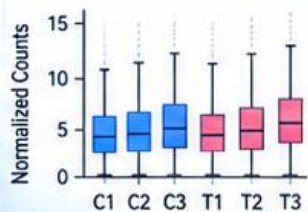
- Normalize counts
- Identify DEGs
- Estimate fold change
- Statistical significance



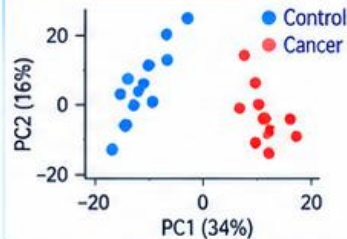
Results (DEGs)

- ↑ Upregulated genes
- ↓ Downregulated genes
- Log2 fold change
- Adjusted p-value (FDR)
- Significant gene list

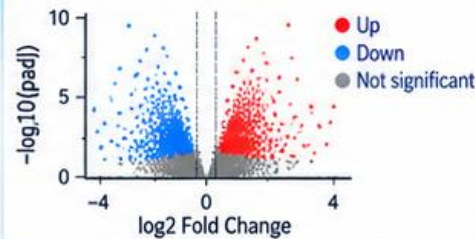
Normalization



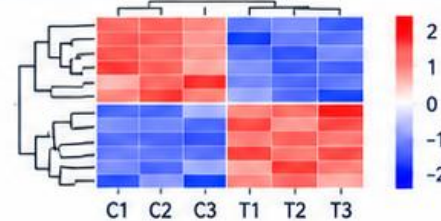
Sample QC (PCA)



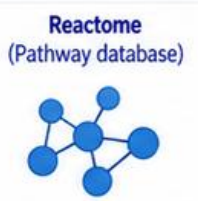
Volcano Plot



Heatmap



Functional Enrichment



LEARNING OBJECTIVES

By the end of this session, you will be able to:



Understand SAM/BAM processing

Learn the structure, format, and information content of SAM/BAM files and their role in RNA-seq analysis.



Convert, sort, and index BAM files

Use samtools to convert SAM to BAM, sort alignments, and create index files for downstream analysis.



Evaluate alignment quality

Assess mapping rate, uniquely/multimapped reads, insert size distribution, and other key QC metrics.



Quantify gene expression with featureCounts

Count reads at the gene level using featureCounts with GTF annotation.



Generate and normalize count matrices

Create a gene $\times$ sample count matrix and normalize counts (e.g., DESeq2 size factors, TPM/CPM).



Perform PCA and sample-level QC

Assess sample similarity, detect outliers, and evaluate data quality using PCA and related QC plots.



Identify differentially expressed genes (DEGs) using DESeq2

Model count data, perform statistical testing, and identify significant DEGs between conditions.



Visualize DEGs using volcano plots and heatmaps

Create publication-quality plots to interpret expression changes and sample clustering.



Interpret biological results

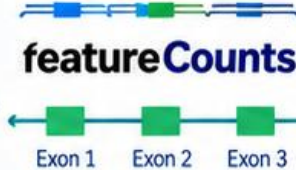
Draw biological meaning from DEGs and connect results to pathways and disease mechanisms.

From Counts
to Knowledge
Explore Biology

Aligned Reads
(Sorted BAM)



Gene Quantification
(featureCounts)



Count Matrix
(genes $\times$ samples)

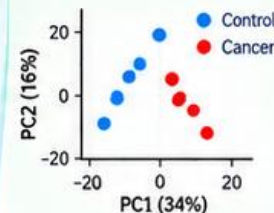
Gene	C1	C2	C3	T1	T2	T3
Gene1	120	95	102	30	28	25
Gene2	0	4	3	50	48	52
Gene3	560	620	580	210	205	198
...	...	...	...	...	...	...

Differential Expression Analysis
(DESeq2)



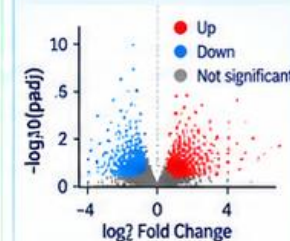
- Normalize counts
- Model count data
- Identify DEGs
- Adjusted p-value (FDR)

Sample QC
(PCA)



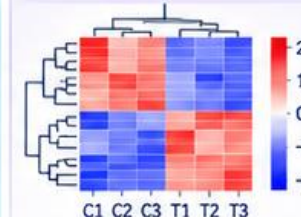
Assess sample similarity
and detect outliers

Volcano Plot



Identify significantly
dysregulated genes

Heatmap



Visualize expression
patterns and sample clustering

Downstream Analysis

Functional Enrichment
(GO / KEGG / Reactome)

Pathway Analysis
(GSEA / GSVA)

PPI Network & Hub Genes

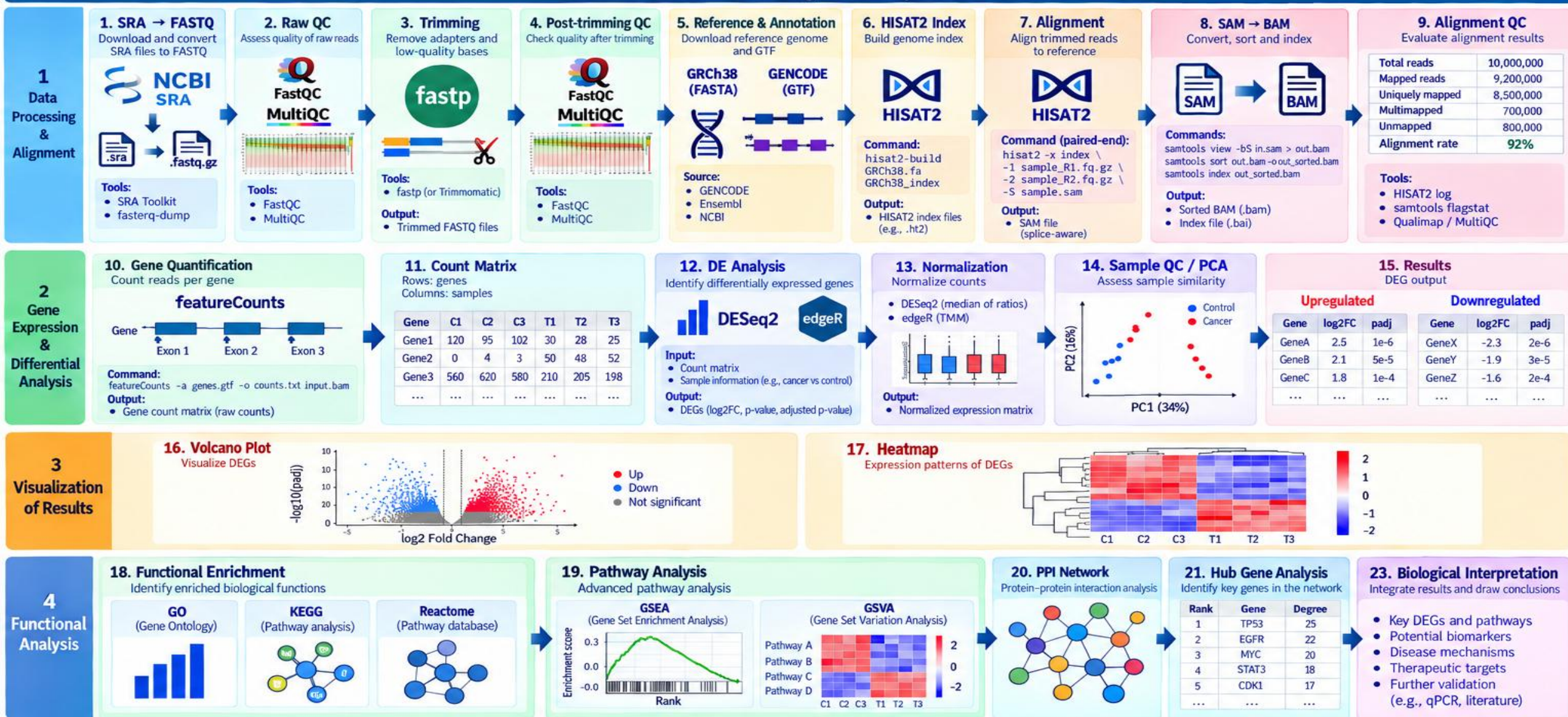
Survival Analysis

Interpret biological meaning
and identify key pathways

Genes to Insights
A Healthier Tomorrow!

Bulk RNA-seq Analysis Workflow (LUAD: 3 Cancer + 3 Control)

From raw data to biological interpretation



Final Output: Comprehensive understanding of molecular mechanisms in LUAD

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Step 9: Understanding SAM

Objective: Understand the information produced by read alignment.

SAM = Sequence Alignment/Map

A text-based format that stores the alignment information of each read to a reference genome.



What does a SAM file contain?

- 1 Read ID** – Unique identifier for each read
- 2 Reference chromosome** – Chromosome where the read aligned
- 3 Mapping position** – Genomic position of alignment
- 4 Alignment flag** – Information about the read (paired/single, strand, etc.)
- 5 MAPQ** – Mapping quality score (confidence of alignment)
- 6 CIGAR string** – Alignment pattern (matches, insertions, deletions, etc.)
- 7 Sequence** – Read nucleotide sequence
- 8 Base quality** – Quality scores for each base
- 9 Paired-end information** – Mate read information (for paired-end data)

Example: SAM file (excerpt)

```
@HD VN:1.6 SO:coordinate
@SQ SN:chr1 LN:248956422
@SQ SN:chr2 LN:242193529
...
```

QNAME	FLAG	RNAME	POS	MAPQ	CIGAR	RNEXT	PNEXT	TLEN	SEQ	QUAL
Cancer1_1	99	chr1	10583	60	101M	=	10673	190	ACTG...	IIIH...
Cancer1_2	147	chr1	10673	60	101M	=	10583	-190	TGCA...	HHHH...
Cancer1_3	83	chr2	34567	50	76M2I23M	=	34678	111	GATC...	IIII...
...										



Importance

SAM records how and where each sequencing read aligned to the reference genome. It contains detailed alignment information, which is essential for downstream analysis (e.g., variant calling, gene quantification).

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Step 10: SAM → BAM Conversion

Objective: Convert the large text-based SAM file into an efficient binary format.



Tool

SAMtools

SAMtools is a widely used toolkit for handling SAM/BAM files.



Command

```
samtools view -@ 18 -b \
alignment/Cancer1.sam \
> bam/Cancer1.bam
```

Conversion Workflow



Text-based
(large file size)
e.g., Cancer1.sam



samtools view



Binary format
(compressed)
e.g., Cancer1.bam

Why convert to BAM?



- ✓ Compressed (smaller file size)
- ✓ Faster to process
- ✓ Efficient storage and I/O
- ✓ Suitable for downstream analysis
- ✓ Can be sorted and indexed

Note:

After conversion, BAM files are typically sorted and indexed for efficient access and downstream analysis.



Key Takeaway

BAM is the preferred format for downstream analysis because it is compressed, smaller, and faster to process compared to SAM.

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Step 11: BAM Sorting

Objective: Arrange aligned reads according to genomic coordinates.



Tool

SAMtools

Used for sorting BAM files based on genomic coordinates.



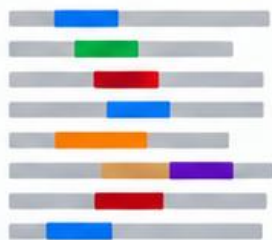
Command

```
samtools sort -@ 18 \
-o bam/Cancer1_sorted.bam \
bam/Cancer1.bam
```

What does sorting do?

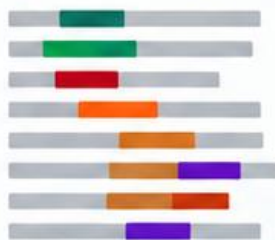
Arranges the aligned reads in the BAM file according to their genomic coordinates.

Unsorted BAM
(random order)



samtools sort

Coordinate-sorted BAM
(genomic order)



Sorting order

Chromosome



Genomic position



Importance

- Coordinate-sorted BAM is required for efficient genomic access and many downstream tools.
- Essential for indexing, visualization, and variant calling.
- Helps in faster data retrieval from specific genomic regions.

Organize Reads
Enable Discoveries

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Step 12: BAM Indexing

Objective: Create an index for rapid access to specific genomic regions.



Tool

SAMtools

Used to create an index file for a sorted BAM file.



Command

```
samtools index -@ 18 \
bam/Cancer1_sorted.bam
```

What does indexing do?

Creates an index file (.bai) that allows tools to quickly access reads from specific genomic regions without scanning the entire file.

Output



Cancer1_sorted.bam



Cancer1_sorted.bam.bai

How it works

Indexed BAM



samtools
index

BAM + Index



Importance

- Allows tools to quickly retrieve reads from selected genomic regions.
- Required for visualization and downstream analysis.
- Improves performance and efficiency.

Used by

- ✓ IGV
- ✓ SAMtools
- ✓ Genome browsers
- ✓ Downstream analysis tools

Indexed for
Faster Insights

Step 13: Alignment QC

Quality Alignments
Reliable Results
Biological Insights

Objective: Determine whether the sequencing reads aligned successfully to the reference genome.

HISAT2 Alignment Metrics

HISAT2 automatically reports the following information:

-  **Total reads**
Number of input reads/pairs
-  **Overall alignment rate**
Percentage of reads aligned
-  **Concordant alignment**
Both paired reads align consistently
-  **Unaligned reads**
Reads that could not be mapped
-  **Multiple alignments**
Reads mapping to multiple genomic locations



Importance

Poor alignment can indicate:

- Poor read quality
- Incorrect reference genome
- Contamination
- Sample-specific problems
- Library preparation problems

View HISAT2 Alignment Statistics



Command (single sample)

```
cat alignment_logs/Cancer1_hisat2.log
```

For all samples

```
cat alignment_logs/Cancer1_hisat2.log  
cat alignment_logs/Cancer2_hisat2.log  
cat alignment_logs/Cancer3_hisat2.log
```

```
cat alignment_logs/Control1_hisat2.log  
cat alignment_logs/Control2_hisat2.log  
cat alignment_logs/Control3_hisat2.log
```

Example HISAT2 Log Output (excerpt)

```
10000000 reads; of these:  
  10000000 (100.00%) were paired; of these:  
    9500000 (95.00%) aligned concordantly 0 times  
    400000 (4.00%) aligned concordantly exactly 1 time  
    100000 (1.00%) aligned concordantly >1 times  
    0 (0.00%) aligned discordantly 1 time  
  500000 (5.00%) aligned 0 times  
Overall alignment rate: 95.00%
```

SAMtools QC



Command (single sample)

```
samtools flagstat bam/Cancer1_sorted.bam
```

Save the report

```
samtools flagstat \  
  bam/Cancer1_sorted.bam \  
  > alignment_logs/Cancer1_flagstat.txt
```

For all samples

```
samtools flagstat bam/Cancer1_sorted.bam > alignment_logs/Cancer1_flagstat.txt  
samtools flagstat bam/Cancer2_sorted.bam > alignment_logs/Cancer2_flagstat.txt  
samtools flagstat bam/Cancer3_sorted.bam > alignment_logs/Cancer3_flagstat.txt
```

```
samtools flagstat bam/Control1_sorted.bam > alignment_logs/Control1_flagstat.txt  
samtools flagstat bam/Control2_sorted.bam > alignment_logs/Control2_flagstat.txt  
samtools flagstat bam/Control3_sorted.bam > alignment_logs/Control3_flagstat.txt
```

Alignment QC Summary



HISAT2 log



Check
alignment metrics



SAMtools flagstat



Evaluate
mapping quality

Good Alignment = Reliable Downstream Analysis

- ✓ Confirms successful read mapping
- ✓ Ensures data quality
- ✓ Helps identify potential issues early
- ✓ Provides confidence for downstream analysis (e.g., quantification, differential expression)

From Alignments
to Discoveries

Gene-Level Quantification

Objective: Determine how many aligned reads are assigned to each gene.

From
Aligned Reads
to Gene Expression

Inputs



Sorted BAM
(aligned reads)



GENCODE v50 GTF
(gene annotation)

Tool



featureCounts
(from Subread package)

Install featureCounts



```
sudo apt update  
sudo apt install -y subread
```

Check installation:

```
featureCounts -v
```

Process Workflow

Sorted BAM
+ GENCODE GTF



featureCounts



Gene-level counts

Command (for all six samples)



```
featureCounts \  
-p  
-T 18  
-a reference/GRCh38.gtf  
-o gene_counts.txt  
bam/*_sorted.bam
```

Key Parameters

Parameter	Meaning
-p	Paired-end sequencing
-T 18	Use 18 CPU threads
-a	GTF annotation file
-o	Output file

Output



gene_counts.txt



gene_counts.txt.summary



Importance

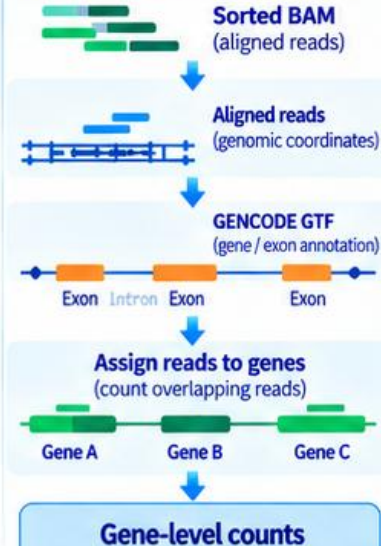
Converts read-level alignment information into gene-level expression counts, which is the basis for downstream analysis (e.g., DESeq2).

How featureCounts Assigns Reads

Objective: Understand how aligned reads are converted into gene counts.

Reads to Genes
Counts to Insights

Process Overview



Example

Gene	Number of reads
Gene A	1,250
Gene B	80
Gene C	600
...	...

Command

```
> featureCounts \  
-p  
-T 18  
-a reference/GRCh38.gtf  
-o gene_counts.txt  
bam/*_sorted.bam
```

Why do we do this?



HISAT2 tells us:
Where did the reads align?



featureCounts tells us:
How many reads are associated with each gene?



Key Takeaway

- featureCounts uses genomic coordinates and GTF annotation to assign aligned reads to genes and generate gene-level counts.
- It converts alignment information into a gene × sample count matrix.

Generate Count Matrix

Objective: Combine gene counts from all six samples into a single matrix.

Multiple Samples
One Expression Matrix

Input (Sorted BAM files)



Cancer1_sorted.bam
Cancer2_sorted.bam
Cancer3_sorted.bam
Control1_sorted.bam
Control2_sorted.bam
Control3_sorted.bam

Command



```
featureCounts \  
-p  
-T 18  
-a reference/GRCh38.gtf  
-o gene_counts.txt  
bam/*_sorted.bam
```

Output



gene_counts.txt
GENE × SAMPLE
COUNT MATRIX

Example (Conceptual)

Gene	Cancer1	Cancer2	Cancer3	Control1	Control2	Control3
Gene A	1250	1370	1310	520	490	550
Gene B	80	95	88	420	390	450
Gene C	600	580	630	590	610	570
...	...	...	...	...	...	...



Importance

The count matrix is the starting input for downstream statistical analysis such as DESeq2.

Examine Count Matrix

Objective: Check whether featureCounts generated the expected output.

Check Your Data
Ensure Reliability

Useful Commands



View first lines
head gene_counts.txt

View summary
cat gene_counts.txt.summary

Count number of genes
grep -v "^#" gene_counts.txt | wc -l

Check file size
ls -lh gene_counts.txt

featureCounts Output Structure

Geneid	Chr	Start	End	Strand	Length	Cancer1	Cancer2	Cancer3	Control1	Control2	Control3
ENSG000001	chr1	11869	14409	+	2541	1250	1370	1310	520	490	550
ENSG000002	chr1	29554	31109	-	1556	80	95	88	420	390	450
ENSG000003	chr1	69091	70008	+	918	600	580	630	590	610	570
...	...	...	...	...	...	...	...	...	...	...	...

Why do we check?

- Ensure all six samples are present
- Gene IDs are present
- Counts are numeric
- Count file is not empty
- Assignment statistics are reasonable



Key Takeaway

Verify the count matrix before proceeding to downstream analysis to ensure reliable results.

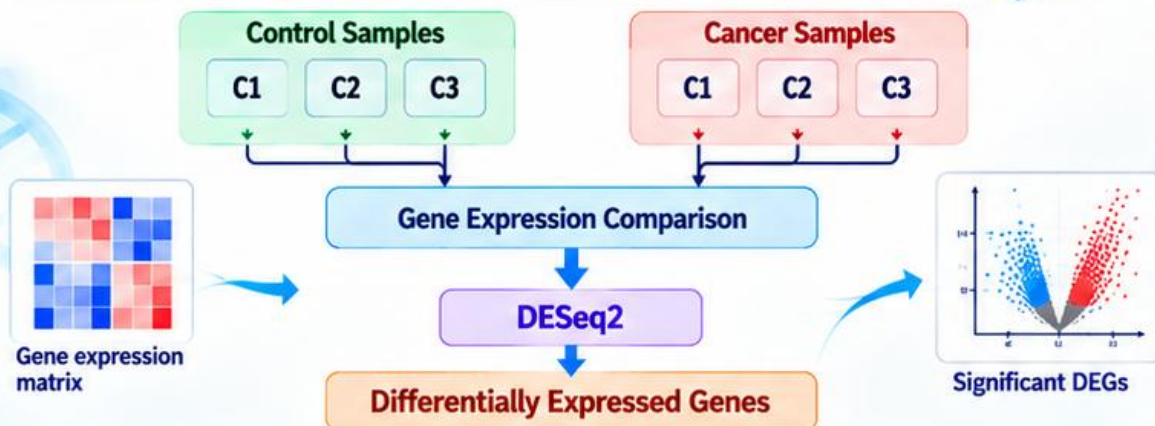
Differential Gene Expression (DGE)



Objective

Identify genes whose expression levels differ significantly between biological conditions.

From Expression Data to Biological Insights



Key Message

DGE analysis reveals genes that are significantly upregulated or downregulated between Control and Cancer samples.

What is Differential Gene Expression?

Same Genes Different Stories



Basic Concept

Differential gene expression analysis compares gene expression between experimental conditions.

It identifies:



Upregulated genes

Increased expression in one condition (e.g., higher in cancer)



Downregulated genes

Decreased expression in one condition (e.g., lower in cancer)



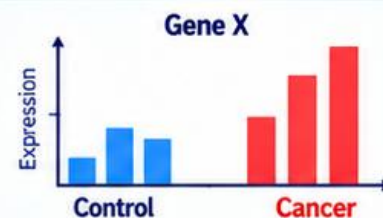
Significant DEGs

Statistically supported expression changes (using adjusted p-value and fold change)



Biological question

"Which genes behave differently between Control and Cancer?"



Example: Gene X is upregulated in cancer

Why Perform DGE Analysis?



Purpose

DGE analysis helps identify genes associated with biological differences between conditions.

Discover Genes Understand Biology Drive Discovery

Applications



Disease-associated genes

Identify genes involved in disease



Altered biological processes

Discover affected pathways and functions



Activated or suppressed pathways

Understand molecular mechanisms



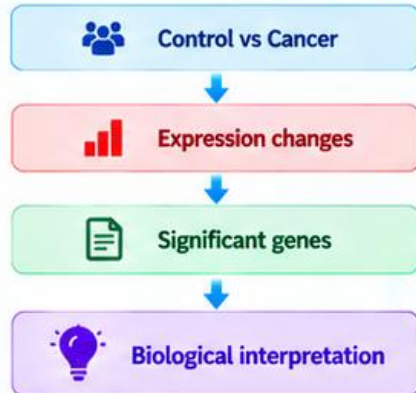
Potential biomarkers

Find diagnostic or prognostic biomarkers



Candidate genes for further investigation

Select targets for validation studies



Input for DESeq2

Right Input Reliable Results

Required Inputs

1. Gene count matrix (Gene × Sample × sample)



- Raw counts from featureCounts
- Rows: genes (e.g., gene IDs)
- Columns: samples
- Values: integer counts

2. Sample metadata (design file)



- Sample names
- Condition information (Control or Cancer)
- Used to define the experimental design in DESeq2



Key Point

DESeq2 requires a gene count matrix and a sample metadata file to perform differential expression analysis.

Example: Count Matrix (excerpt)

Gene	C1	C2	C3	Cancer1	Cancer2	Cancer3
GeneA	1250	1370	1310	520	490	550
GeneB	80	95	88	420	390	450
GeneC	600	580	630	590	610	570
...	...	...	...	...	...	...

Example: Sample Metadata

Sample	Condition
Control1	Control
Control2	Control
Control3	Control
Cancer1	Cancer
Cancer2	Cancer
Cancer3	Cancer

DESeq2 Package



DESeq2 is an R/Bioconductor package designed for differential expression analysis of RNA-seq count data.

For
Reliable
Differential
Expression
Analysis

Main Features



Normalizes for sequencing-depth differences



Estimates biological variability



Models count data using a negative binomial distribution



Performs statistical testing



Corrects for multiple testing (e.g., Benjamini-Hochberg FDR)



Key Message

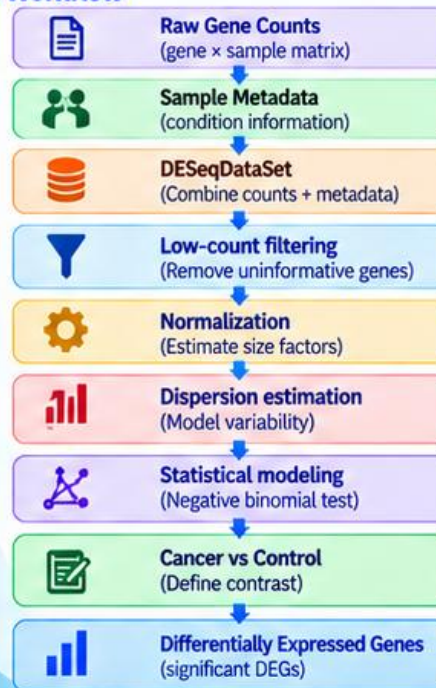
DESeq2 provides a robust and widely used framework for identifying differentially expressed genes from count data.



R / Bioconductor
For RNA-seq
Differential Expression

DESeq2 Analysis Workflow

Overall Workflow



From
Counts
to
Biological
Insights

Create DESeq2 Dataset

R Code



```
library(DESeq2)

dds <- DESeqDataSetFromMatrix(
  countData = count_matrix,
  colData = colData,
  design = ~ condition
)
```

Combine
Your Data
for Analysis



DESeqDataSet
Ready for
Downstream Analysis

Key Concept

design = ~ condition

tells DESeq2 that condition is the experimental factor being tested.

What is included?

- ✓ countData : gene x sample count matrix
- ✓ colData : sample metadata (e.g., Control or Cancer)
- ✓ design : experimental design formula

Define the Reference Condition

Control as Reference

For a Control vs Cancer experiment:

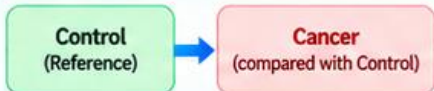


```
dds$condition <- relevel(
  dds$condition,
  ref = "Control"
)
```

Why?

- ✓ The reference determines how the fold change is interpreted.
- ✓ A positive log2 fold change means higher expression in Cancer relative to Control.
- ✓ Ensures correct direction of up- and downregulation.

Concept



Key Point

Always set the biologically meaningful reference condition (e.g., Control).

Set the Right
Comparison

Filter Low-Count Genes

Why Filter?

- ✓ Genes with extremely low read counts provide limited statistical information.
- ✓ Can lead to unreliable estimates of dispersion and fold change.
- ✓ Filtering improves the power and accuracy of DE analysis.

R Code

```
keep <- rowSums(counts(dds) >= 10) >= 3
dds <- dds[keep, ]
```

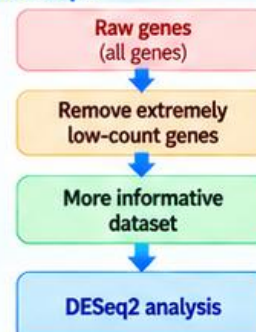


Typical filtering criterion

Keep genes with at least 10 counts in 3 or more samples (adjustable).

Keep
What Matters

Concept



Run DESeq2

Statistical Analysis



```
dds <- DESeq(dds)
```

DESeq2 performs:

- ✓ Normalization (estimate size factors)
- ✓ Dispersion estimation (model biological variability)
- ✓ Negative binomial modeling (for count data)
- ✓ Hypothesis testing (Wald test)
- ✓ Multiple-testing correction (Benjamini-Hochberg FDR)

What Happens Internally?



Key Message

A single command (DESeq) performs normalization, dispersion estimation, modeling, testing, and multiple-testing correction.

Let the Statistics
Do the Work

1. Why RNA-seq Counts Need Statistical Modeling

RNA-seq produces integer read counts



For gene i in sample j :

K_{ij} = number of reads assigned to gene i

The count depends on:

- True biological expression
- Sequencing depth (library size)
- Biological variability (between replicates)
- Technical variation (sequencing, library prep, etc.)

Therefore:

Observed Count $\neq$ True Expression



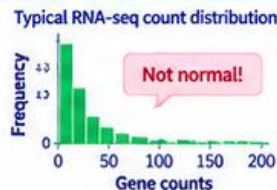
Key Message

DESeq2 uses a statistical model to separate biological signal from technical and sampling variation.

2. Why Not Use a Simple t-test?

RNA-seq counts are not normally distributed

- Discrete (integer values)
- Non-negative
- Often highly variable
- Mean-dependent variance
- Typically right-skewed



A simple normal model is inappropriate:

$$K \sim N(\mu, \sigma^2) \quad \text{X Not suitable for RNA-seq counts}$$

DESeq2 uses the Negative Binomial distribution:

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i)$$

μ_{ij} = expected count for gene i in sample j

α_i = dispersion of gene i



Key Message

The negative binomial model accounts for the extra variability (overdispersion) in RNA-seq data.

3. Negative Binomial Model

Core DESeq2 model

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i) \quad \text{For gene } i \text{ and sample } j:$$

Mean and variance

$$E(K_{ij}) = \mu_{ij}$$

$$\text{Var}(K_{ij}) = \mu_{ij} + \alpha_i \mu_{ij}^2$$

Two components of variance

Variance =
Poisson variation
(μ)

+

Biological overdispersion
($\alpha \mu^2$)

Key points

- ✓ When $\alpha = 0$, the model reduces to the Poisson distribution.
- ✓ The dispersion parameter (α) allows for biological variability between replicates.
- ✓ Each gene has its own dispersion estimate.



Key Message

The negative binomial model provides a flexible framework to model the mean-variance relationship in RNA-seq data.

4. Library Size / Size Factor

Different samples can have different sequencing depths

Sample	Total reads (millions)
Control 1 (C1)	20
Control 2 (C2)	30
Control 3 (C3)	22
Cancer 1 (T1)	45
Cancer 2 (T2)	38
Cancer 3 (T3)	40

Samples with more reads will naturally have larger counts.

DESeq2 estimates a size factor s_j for each sample.

The expected count is modeled as:

$$\mu_{ij} = s_j q_{ij}$$

s_j = sample-specific size factor
 q_{ij} = underlying expression level for gene i in sample j

Observed count $\approx$ Library size $\times$ Biological expression



Key Message

Size factors correct for differences in sequencing depth so that samples become comparable.

5. Median-of-Ratios Normalization

DESeq2 normalization algorithm

- 1 Compute the **geometric mean** for each gene (across all samples):

$$g_i = \left(\prod_j K_{ij} \right)^{1/m} \quad \begin{array}{l} g_i = \text{geometric mean for gene } i \\ m = \text{number of samples} \end{array}$$

- 2 Compute the ratio for each sample:

$$r_{ij} = \frac{K_{ij}}{g_i}$$

- 3 Estimate the size factor for each sample:

$$s_j = \text{median}_i(r_{ij})$$

Genes with zero counts across all samples are excluded.



Key Message

The median-of-ratios method makes samples comparable despite differences in sequencing depth and composition.

6. Normalized Counts

Once size factors are estimated:

$$K_{ij}^{\text{normalized}} = \frac{K_{ij}}{s_j}$$

Example

Raw count (K_{ij}) = 1000

Size factor (s_j) = 2

Normalized count = $1000 / 2 = 500$

Important points

- ✓ Normalized counts are useful for visualization and interpretation (e.g., PCA, heatmaps).
- ✓ DESeq2 does not simply perform a t-test on normalized counts. The size factor is incorporated directly into the statistical model.



Key Message

Normalization removes sequencing depth effects, but statistical testing is performed within the negative binomial model using the size factors.

7. Modeling Experimental Conditions

Example experiment

Control (n = 3)

C1

C2

C3

Cancer (n = 3)

T1

T2

T3

Model for expected count

$$\mu_{ij} = s_j q_{ij}$$

where q_{ij} is modeled using a generalized linear model (GLM).

For two conditions (Control vs Cancer):

$$\log(q_{ij}) = \beta_0 + \beta_1 X_j$$

$X_j = 0$ for Control
 $X_j = 1$ for Cancer

Parameters

- β_0 = baseline expression (Control)
- β_1 = log2 fold change for Cancer vs Control
- X_j = condition indicator variable



Key Message

DESeq2 uses a GLM within a negative binomial framework to estimate condition effects while accounting for variability and library size.

8. From Model to Differential Expression

Testing for differential expression

Hypothesis test for each gene:

$H_0 : \beta_1 = 0$ (no difference)

$H_1 : \beta_1 \neq 0$ (differential expression)

Estimated log2 fold change

$$\log_2 FC = \hat{\beta}_1 / \log(2)$$

Statistical significance

- Wald test is used to test the hypothesis.
- Raw p-values are computed for each gene.
- Multiple testing correction (Benjamini-Hochberg FDR).

Final result

Genes with $\text{padj} < 0.05$ are considered differentially expressed (often with an additional threshold $|\log_2 FC| \geq 1$).



Key Message

The statistical model allows us to identify genes that show significant expression changes between Control and Cancer, while properly accounting for sequencing depth and biological variability.

Control vs Cancer Comparison



Objective

Define the contrast to compare gene expression between Cancer and Control samples.

Compare Conditions
Discover Key Genes

R Code



```
res <- results(
  dds,
  contrast = c("condition",
    "Cancer", "Control"))
)
```

Comparison

Cancer
(n = 3)

vs

Control
(n = 3)

Differential Expression
Results

What does it do?

- ✓ Specifies the experimental contrast (Cancer vs Control).
- ✓ Calculates the estimated expression difference between the two conditions.
- ✓ Generates a results table with log2 fold change, p-value and adjusted p-value.

Understanding log2 Fold Change



log2FoldChange

Measures the magnitude and direction of gene-expression change between Cancer and Control.

Direction
and Magnitude
of Change

log2FC	Interpretation
+2	4× higher in Cancer
+1	2× higher in Cancer
0	No change
-1	2× lower in Cancer
-2	4× lower in Cancer



Positive log2FC
→ Upregulated
(higher in Cancer)



Negative log2FC
→ Downregulated
(lower in Cancer)



Key Message

The log2 fold change indicates both the magnitude (how much) and the direction (up or down) of gene expression change.

Understanding Statistical Significance



Adjusted p-value

Thousands of genes are tested simultaneously. Therefore, multiple-testing correction is required.

Control
False Discoveries
Increase Confidence

pvalue
(raw significance)

Multiple-testing
correction
(e.g., Benjamini-Hochberg)

padj
(adjusted p-value)

DESeq2 provides

- ✓ pvalue — raw p-value from statistical test
- ✓ padj — p-value adjusted for multiple testing (False Discovery Rate)

Common threshold

- ✓ **padj < 0.05**
indicates statistically significant differential expression after correction.



Key Message

Adjusted p-values (padj) control the false discovery rate and help identify truly significant genes.

Identifying Differentially Expressed Genes



Common DEG Criteria

Select genes based on adjusted p-value and fold change threshold.

Filter
Find Key Players

R Code



```
sig <- subset(
  as.data.frame(res),
  padj < 0.05 &
  abs(log2FoldChange) >= 1
)
```

Classification of DEGs

DEGs

(padj < 0.05 & |log2FC| ≥ 1)

log2FC ≥ 1
→ Upregulated
(higher in Cancer)

log2FC ≤ -1
→ Downregulated
(lower in Cancer)

What do we get?

- ✓ A list of statistically significant differentially expressed genes.
- ✓ Genes are classified as upregulated or downregulated based on log2FC.
- ✓ These genes can be used for downstream functional analysis.

DESeq2 Results



Important Columns

The results table contains key statistics for each gene.

From
Numbers to
Biological Meaning

Column	Meaning
baseMean	Mean normalized expression
log2FoldChange	Estimated expression change (Cancer vs Control)
lfcSE	Standard error of the log2 fold change
stat	Wald test statistic
pvalue	Raw p-value
padj	Adjusted p-value (multiple testing correction)

Example (first few rows)

gene	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
GeneA	125.3	2.14	0.32	6.69	2.1e-11	4.3e-08
GeneB	56.7	-1.25	0.28	-4.52	6.1e-06	8.7e-04
GeneC	892.1	0.05	0.41	0.12	0.90	0.98
...	...	...	...	...	...	...



Key Message

Each row represents a gene and its statistical evidence for differential expression.

Summary: From DEGs to Insights



Key Takeaways

DESeq2 helps identify genes that are significantly upregulated or downregulated between Control and Cancer.

Same Data
New Biological
Insights

Count Matrix
(Gene × Sample)

DESeq2 Analysis
(Cancer vs Control)

Significant DEGs
(padj < 0.05 & |log2FC| ≥ 1)

Biological Interpretation
(Pathways, Functions, Biomarkers)

What We Have Learned

- ✓ How to define the contrast (Cancer vs Control)
- ✓ How to interpret log2 fold change
- ✓ Why adjusted p-values are important
- ✓ How to identify significant DEGs
- ✓ What the results table contains
- ✓ These DEGs can be used for functional enrichment, pathway analysis and biomarker discovery.



Goal

Turn gene expression differences into meaningful biological insights to understand cancer biology.

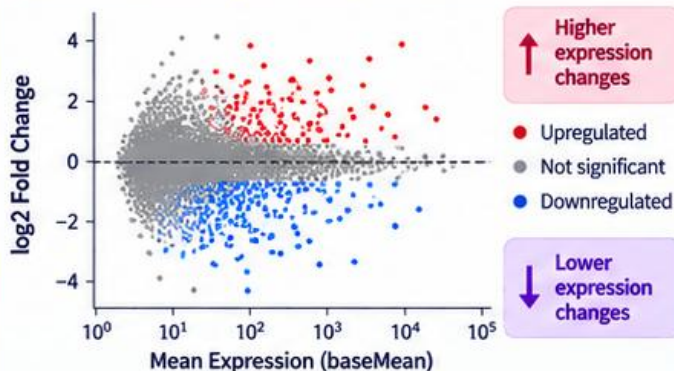
MA Plot

Visualizing Expression Changes



The MA plot displays:

- X-axis: Mean expression (average normalized counts)
- Y-axis: log2 fold change



Purpose

Helps visualize the distribution of expression changes across all genes.

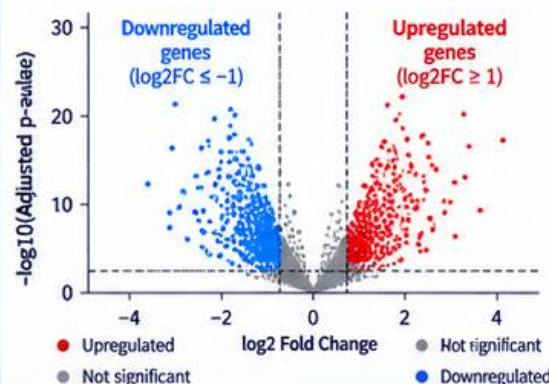
Volcano Plot

Fold Change vs Statistical Significance



The volcano plot combines:

- Fold change (log2FC)
- Statistical significance (adjusted p-value)



Interpretation

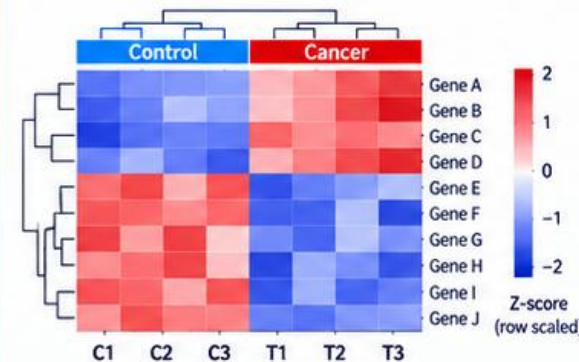
- Right side → Upregulated genes
- Left side → Downregulated genes
- Higher → More statistically significant genes

Heatmap of DEGs

Expression Pattern Across Samples



A heatmap shows expression patterns of significant genes across all samples.



Purpose

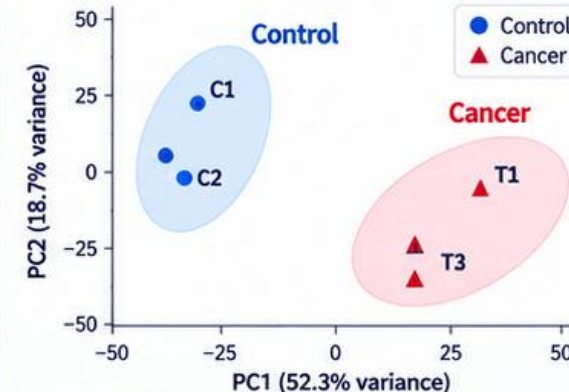
Determine whether samples and biological groups show consistent expression patterns.

PCA Plot

Sample Clustering and Variation



PCA evaluates overall similarity between samples.



Purpose

- Detect sample clustering
- Assess biological variation
- Identify potential outliers
- Evaluate separation between conditions.

From DEGs to Biological Meaning

Functional Interpretation



Differentially Expressed Genes
(significant up and downregulated genes)



Gene Annotation
(Map to gene IDs, symbols, functions)



GO Enrichment
(Biological processes, cellular components, molecular functions)



KEGG Pathways
(Pathway enrichment analysis)



Biological Processes and Pathways



Interpretation
Understand the biological meaning and generate hypotheses

Applications

- ✓ Identify key biological processes
- ✓ Discover affected pathways
- ✓ Find potential biomarkers
- ✓ Generate hypotheses for further validation

Common Tools

- clusterProfiler (R)
- DAVID
- Enrichr
- g:Profiler
- KEGG Mapper

Complete RNA-seq → DGE Workflow

From Sequencing to Biological Insights

Raw Data Processing



FASTQ (Raw sequencing data)



Quality Control (FastQC, MultiQC)



Trimming (Trimmomatic / fastp)

Alignment and Quantification



Alignment (HISAT2)



Sorted BAM (SAMtools)



Gene-level Counting (featureCounts)

Differential Expression



Gene Count Matrix
(genes × samples)



DESeq2 (Control vs Cancer)



Differentially Expressed Genes
(Upregulated / Downregulated)

Functional Interpretation



GO / KEGG Analysis



Biological Interpretation

Key Takeaways

Differential Gene Expression Analysis

01

DGE identifies genes with different expression between conditions.

02

DESeq2 is designed for raw RNA-seq count data.

03

Experimental design and reference group are critical.

04

log2FC describes the direction and magnitude of change.

05

padj accounts for multiple testing.

06

PCA and heatmaps help assess sample-level patterns.

07

Volcano and MA plots visualize differential expression.

08

DEGs can be taken forward for GO, KEGG, pathway, and network analysis.



Final Message

From raw sequencing data to biological interpretation, differential gene expression analysis helps us uncover the molecular mechanisms underlying disease, such as cancer.

Data
→ Genes
→ Biology
→ Impact

1. Modeling Experimental Conditions

Experimental design

Control (n=3)
C1 C2 C3

Cancer (n=3)
T1 T2 T3

DESeq2 models the expected count as:

$$\mu_{ij} = s_j q_{ij}$$

The biological effect is represented using a generalized linear model (GLM).

For two conditions:

$$\log(q_{ij}) = \beta_0 + \beta_1 X_j$$

β_0 = baseline expression
 β_1 = effect of Cancer
 X_j = condition indicator (0 = Control, 1 = Cancer)

Model gene expression differences between conditions.

2. What Is log2 Fold Change?

Fold change interpretation

$$LFC = \log_2 \left(\frac{\text{Expression}_{\text{Cancer}}}{\text{Expression}_{\text{Control}}} \right)$$

Cancer / Control	log2FC	Interpretation
4	+2	4x higher in Cancer
2	+1	2x higher in Cancer
1	0	No change
0.5	-1	2x lower in Cancer
0.25	-2	4x lower in Cancer

Positive log2FC
→ Upregulated (higher in Cancer)

Negative log2FC
→ Downregulated (lower in Cancer)

Key Message
The log2 fold change indicates both the magnitude and direction of gene expression change.

3. Hypothesis Testing

Test for differential expression

Null hypothesis (H_0)
 $H_0 : \beta_1 = 0$
There is no difference between Control and Cancer.

Alternative hypothesis (H_A)
 $H_A : \beta_1 \neq 0$
There is a difference between Control and Cancer.

Test statistic (approximately)
$$z = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

 $\hat{\beta}_1$ = estimated log2 fold-change effect
 SE = standard error

↓

p-value
(obtained from the test statistic)

Key Message
DESeq2 tests each gene to determine whether the observed expression difference is statistically significant.

4. Multiple Testing Correction

Why adjust p-values?

- ✓ In RNA-seq, thousands of genes are tested simultaneously (e.g., ~20,000 genes).
- ✓ Using a simple threshold ($p < 0.05$) would result in many false positives purely by chance.
- ✓ DESeq2 adjusts p-values using the Benjamini-Hochberg (BH) procedure.

Goal: Control the False Discovery Rate (FDR)

Adjusted p-value is reported as:

padj

Reduce false positives and increase confidence in results.

5. Benjamini-Hochberg Concept

BH adjustment procedure
Suppose genes have ordered p-values:

$$p_1 \leq p_2 \leq \dots \leq p_m$$

For the i^{th} ranked gene:

$$p_i^{BH} = \frac{p_i m}{i}$$

m = total number of tests
 i = rank of the p-value

The values are then adjusted to maintain monotonicity (smaller p-values remain smaller).

Interpretation
padj < 0.05 means the gene passes the selected FDR threshold.

Control the False Discovery Rate (FDR)

6. DEG Selection

Common criteria for identifying DEGs
A gene is considered differentially expressed if:

padj < 0.05 (statistical significance) and **$|\log_2 FC| \geq 1$** (biological significance)

Upregulated genes
 $\log_2 FC \geq 1$ & padj < 0.05

Downregulated genes
 $\log_2 FC \leq -1$ & padj < 0.05

Key Message
Combining adjusted p-value and fold change helps identify biologically meaningful genes.

7. DESeq2 Algorithm: Overall

Simplified workflow

- 1 Raw count matrix (genes x samples)
- 2 Remove very low-count genes
- 3 Calculate size factors (median-of-ratios)
- 4 Normalize for sequencing depth
- 5 Estimate gene-wise dispersion
- 6 Fit dispersion trend
- 7 Shrink/estimate dispersion
- 8 Fit Negative Binomial GLM
- 9 Estimate log2 fold change
- 10 Hypothesis testing
- 11 Benjamini-Hochberg correction
- 12 Differentially Expressed Genes (DEGs)

From raw counts to meaningful results.

8. The Mathematical Model in One Slide

DESeq2 model
For gene i and sample j :

$$K_{ij} \sim NB(\mu_{ij}, \alpha_i)$$

with:

$$\mu_{ij} = s_j q_{ij}$$

and:

$$\log(q_{ij}) = X_j \beta_i$$

Therefore:

$$K_{ij} \sim NB(s_j \exp(X_j \beta_i), \alpha_i)$$

K_{ij} = observed read count
 s_j = size factor
 X_j = experimental design
 β_i = gene-specific coefficients
 α_i = gene-specific dispersion

This is the mathematical core of DESeq2.

9. Example: Control vs Cancer

Example gene

Control (n=3)
C1 C2 C3

Cancer (n=3)
T1 T2 T3

Counts

Condition	C1	C2	C3
Control	100	110	90
Cancer	400	450	420

Model estimates:

Cancer / Control ≈ 4

$\log_2 FC \approx 2$

$p = 0.0001$

padj = 0.002

Conclusion
 $\log_2 FC \approx +2$, padj = 0.002
→ Significantly upregulated in Cancer relative to Control.

10. Visualizing DESeq2 Results

MA Plot
log2FC vs Mean expression

Volcano Plot
log2FC vs $-\log_{10}(\text{padj})$

Heatmap (Top DEGs)

Legend: ● Upregulated, ● Not significant, ● Downregulated

11. PCA Plot

Sample clustering

PCA (52.3% variance)

Legend: ● Control, ● Cancer

PCA helps assess sample clustering, detect outliers, and evaluate separation between conditions.

12. From DEGs to Biological Meaning

Functional interpretation

- Differentially Expressed Genes (Upregulated / Downregulated)
- Gene Annotation (Map to gene IDs, symbols, functions)
- GO Enrichment (Biological processes, cellular components, molecular functions)
- KEGG Pathways (Pathway enrichment analysis)
- Biological Interpretation (Understand biological mechanisms)

From genes to biological insights.

13. Complete RNA-seq → DGE Workflow

FASTQ (Sequencing data) → (LQC, MultiQC) → (Trimmomatic/fastp) → Alignment (HISAT2) → Sorted BAM (SAMtools) → Sorted BAM (SAMtools) → featureCounts (Gene counts) → Count Matrix (genes x samples) → DESeq2 (Control vs Cancer) → DEGs (Up / Down) → GO / KEGG Biological interpretation

14. Why DESeq2 Works

- 1 Models count data using the negative binomial distribution ($\mu + \alpha\mu^2$).
- 2 Accounts for sequencing depth using size factors.
- 3 Estimates biological variability (dispersion) across replicates.
- 4 Uses a robust statistical framework with FDR correction.

15. Key Mathematical Concepts

Concept	Purpose
Raw counts (K_{ij})	Starting data (integer read counts)
Size factor (s_j)	Correct sequencing-depth differences
Negative Binomial (NB)	Model RNA-seq count variability ($\mu + \alpha\mu^2$)
Dispersion (α_i)	Model biological variability between replicates
log2 Fold Change (log2FC)	Measure expression change (direction and magnitude)
BH-adjusted p-value (padj)	Control false discovery rate (FDR)

16. Key Takeaways

- ✓ DESeq2 uses a negative binomial model to analyze RNA-seq count data.
- ✓ Size factors correct for differences in sequencing depth.
- ✓ Dispersion models biological variability.
- ✓ log2FC indicates the direction and magnitude of expression change.
- ✓ Adjusted p-values (padj) control the false discovery rate.
- ✓ PCA, heatmaps, MA and volcano plots help visualize results.
- ✓ DEGs can be used for GO, KEGG, pathway and network analysis.

From statistical modeling to biological discovery.

Genes → Functions → Insights